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B737 FLEET MAGAZINE

A 20 page exclusive fleet magazine for EAL 737 pilots.

Fleet News • Deeper look into B737 incidents and accidents • Breakthroughs in flight • Notices • System reviews

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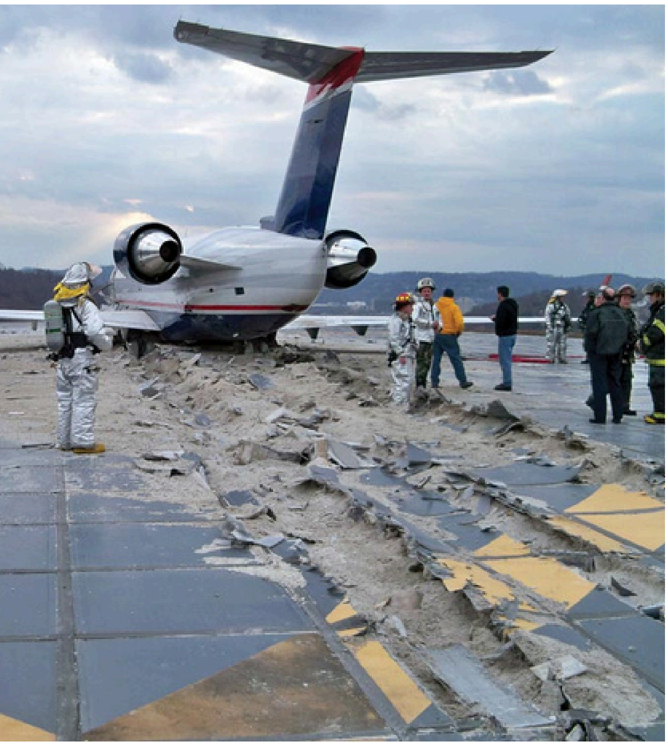


A NEW ERA IN AIRPORT PAVEMENT MANAGEMENT

ACR/PCR

By Cpt Israel Ya

1.0 Definitions

1.1 ACR - Aircraft Classification Rating is a number expressing the relative effect of an aircraft on a pavement for a specified standard subgrade strength.

1.2 PCR - Pavement Classification Rating is a number expressing the bearing strength of a pavement for unrestricted operations.

2.0 Introduction

Adopting the ACR-PCR system requires a comprehensive understanding of the new system's requirements for all operators as this new method is mandatory for all airports and aerodromes within the scope of EASA and ICAO.

An airport payment must be capable of

- Satisfying bearing strength capabilities to accommodate an analyzed aircraft mix (Vertical)

- Providing sufficient friction characteristics to maintain aircraft under control in critical phases(horizontal)

A Pavement Rating System Classifies the pavements with regard to their structural capacity of sustaining a forecasted aircraft mix, based on two elements associated to the PAVEMENT and AIRCRAFT.

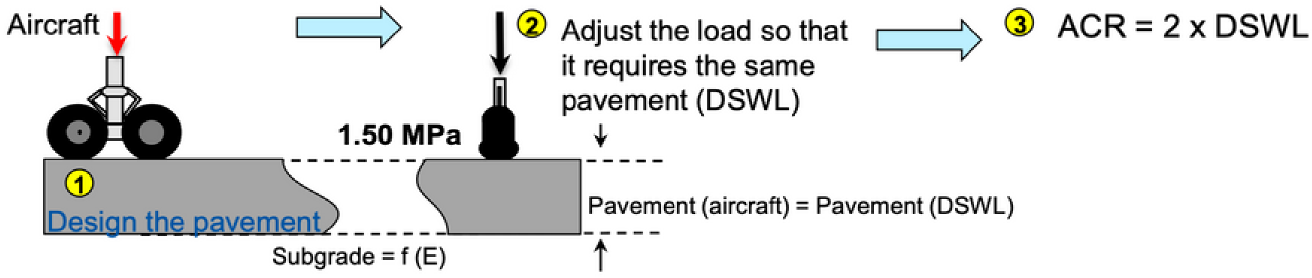
The ACR-PCR system is structured so a pavement with a particular PCR value can support an aircraft that has an ACR value equal to or less than the pavement's PCR value. This is possible because ACR and PCR values are computed using the same technical basis.

- If **ACR ≤ PCR**, the aircraft can operate on the pavement without restriction,
- If **ACR > PCR,** the aircraft may be excluded, or may be allowed to operate subject to weight and/or frequency limitations

3.0 Why The Change?

The **ACN – PCN** system is

- Unable to consider accurately "complex" landing gear configurations
- Doesn't account for the improved characteristics of new-generation pavement materials
- Inconsistent with modern pavement design methods
- Unable to consider the variability of landing gear transverse positions (different overall wheel tracks)
- Lack of guidance for PCN determination



4.0 THE ACR CONCEPT

Similarly to the ACN, the ACR is computed as twice the Derived Single Wheel Load (DSWL), which is the load on a single, isolated wheel (with fixed tire pressure) requiring the same pavement thickness as the considered aircraft

The changes from ACN are:

1. The pavement is designed according to a rational pavement design procedure (vs. CBR or Westergaard procedures): this is the major change vs. ACN and the key part of ACR computation
2. The DSWL is computed for a tire pressure of 1.50 MPa (vs. 1.25 MPa)
3. The ACR is expressed in hundreds of kilograms (vs. tons)

5.0 THE PCR CONCEPT

The PCR of a pavement should reflect the pavement design with respect to the traffic it is intended to serve

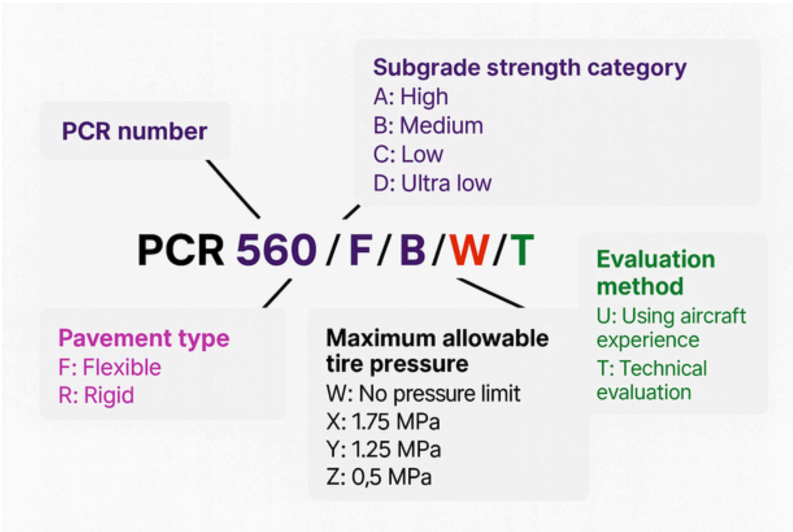
The PCR procedure should ensure that:

- If the pavement CDF is equal to or lower than 1.0 (well or over-designed), no aircraft weight restriction should occur
- If the pavement CDF is higher than 1.0, weight restrictions should apply to one aircraft at least

Unlike PCN a generic PCR computation procedure with several degrees of freedom (e.g. pavement damage model) is proposed

6.0 ACR PCR REPORTING FORMAT

- Similarly to the PCN, the PCR represents the pavement bearing strength (on the ACR scale) for unrestricted operations
- A PCR should be determined by the airport operator for all the pavements intended for aircraft of mass greater than 5.7 tons
- The PCR should be published in the Aeronautical Information Publication (AIP) according to the format defined in ICAO Annex 14 (§ 2.6.6)



THUNDERSTORM TACTICS

Avoidance is the Win

By Cpt Duresa D

As the rainy season approaches in Ethiopia, our skies will begin to fill once again with the towering convective systems that define this time of year. For us as Boeing 737 NG and MAX pilots, this mean renewed focus on weather awareness, decision-making, and radar proficiency.

Flying across Africa offers breathtaking views—and equally impressive thunderstorms. Towering cumulonimbus clouds, rapid weather changes, and seasonal convective systems are routine, especially during the wet season. As 737 pilots, we have the tools and training to navigate these threats—but the best strategy is still: don't engage. Avoid.

SPACING AND DEVIATION:
HOW FAR IS FAR ENOUGH?

BOEING AND ICAO RECOMMEND:

At least 20 NM laterally from thunderstorms at or above FL200
Greater spacing on the downwind side due to Windshear risk, Do not attempt to fly between two cells unless the gap is wide, stable, and radar clear.

DO NOT BE PRESSURED BY ATC OR ROUTE EFFICIENCY

Deviation is justified when safety is at stake. Communicate early and clearly.

WHY AVOIDANCE MATTERS MORE THAN EVER

Modern aircraft are structurally strong, and the 737 is well-equipped. But thunderstorms remain one of the most dangerous weather phenomena in aviation. They bring with them:

- Severe turbulence and windshear
- Hail and lightning strikes
- Microbursts
- Icing and engine flameouts
- Instrument and autopilot anomalies due to static discharge

These aren't just checklist events—they're threats to flight safety that escalate quickly. The best defense? Staying clear from the start.

RADAR: YOUR EARLY WARNING SYSTEM

Both the NG and MAX aircraft have sophisticated weather radar, but its effectiveness depends on how well it's used. Automation doesn't replace pilot scanning. Here's a quick refresher:

Radar Tilt: What You Don't See Can Hurt You

- At cruise, keep a slight down-tilt (around -1˚ to -2˚) to scan just below the flight path.
- During climb or descent, adjust to scan through levels you will pass through.
- Avoid excessive down-tilt at high altitude—it can miss cells building at your level.

Gain Settings: Auto ≠ Always Right

- Use manual gain to confirm the strength of returns.
- High gain can exaggerate returns; low gain can miss embedded CBs.
- Use WX+T (Weather + Turbulence) mode on the MAX when available—it provides better differentiation.

FINAL THOUGHTS - As we head into Ethiopia's rainy season, now is the time to refresh our weather flying skills. With our experience, equipment, and teamwork,
Ethiopian 737 pilots are well-prepared. But don't let confidence lead to complacency. In convective weather, the best pilots are the ones who simply… don't go there.
Avoid. Communicate. Adjust early. Because in our skies, safety is always worth the detour.